

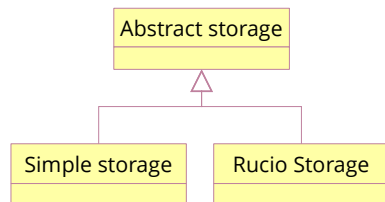


SKAO Regional Centre United Kingdom

IVOA Execution Broker

Progress report

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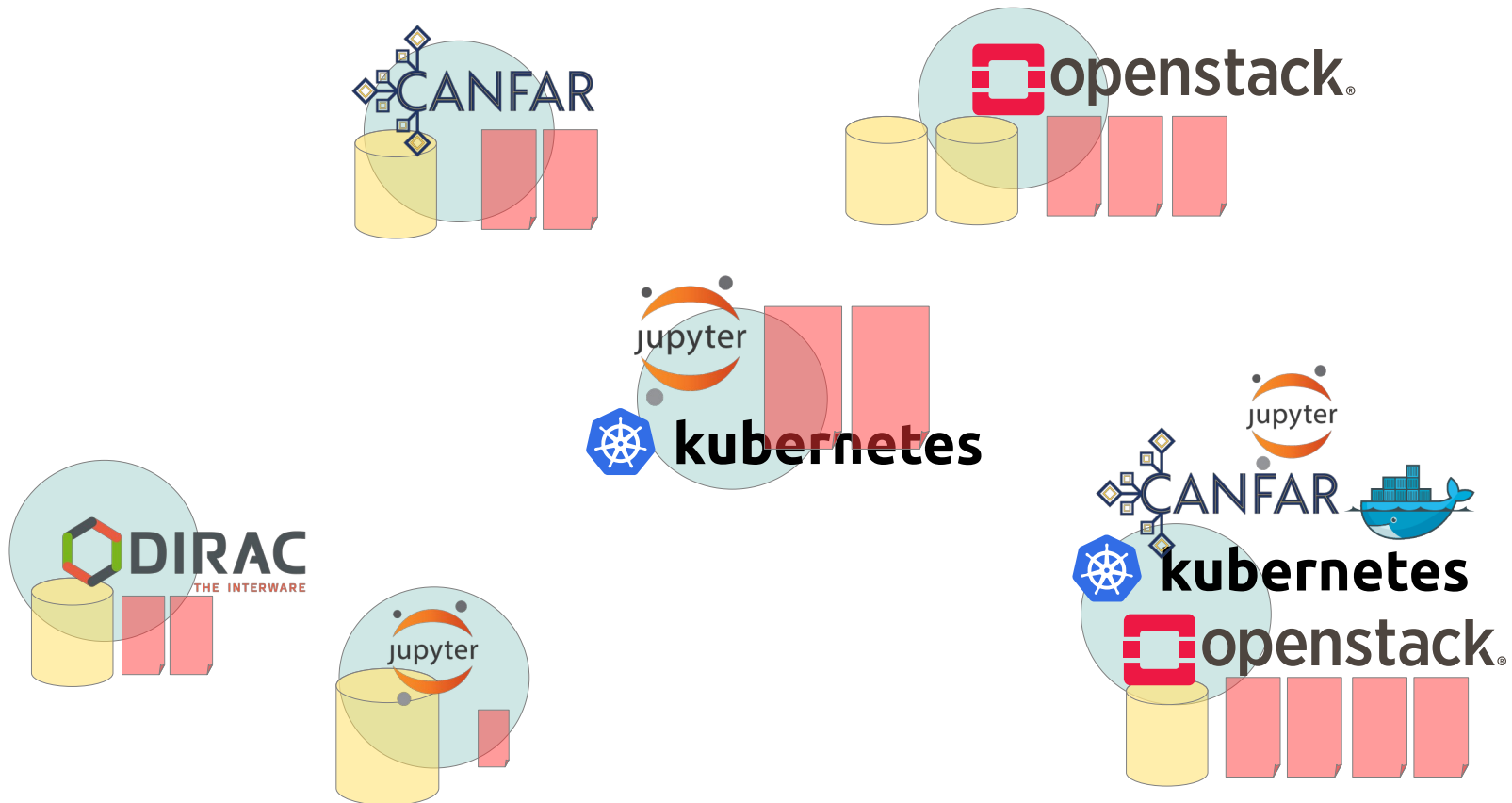
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The problem

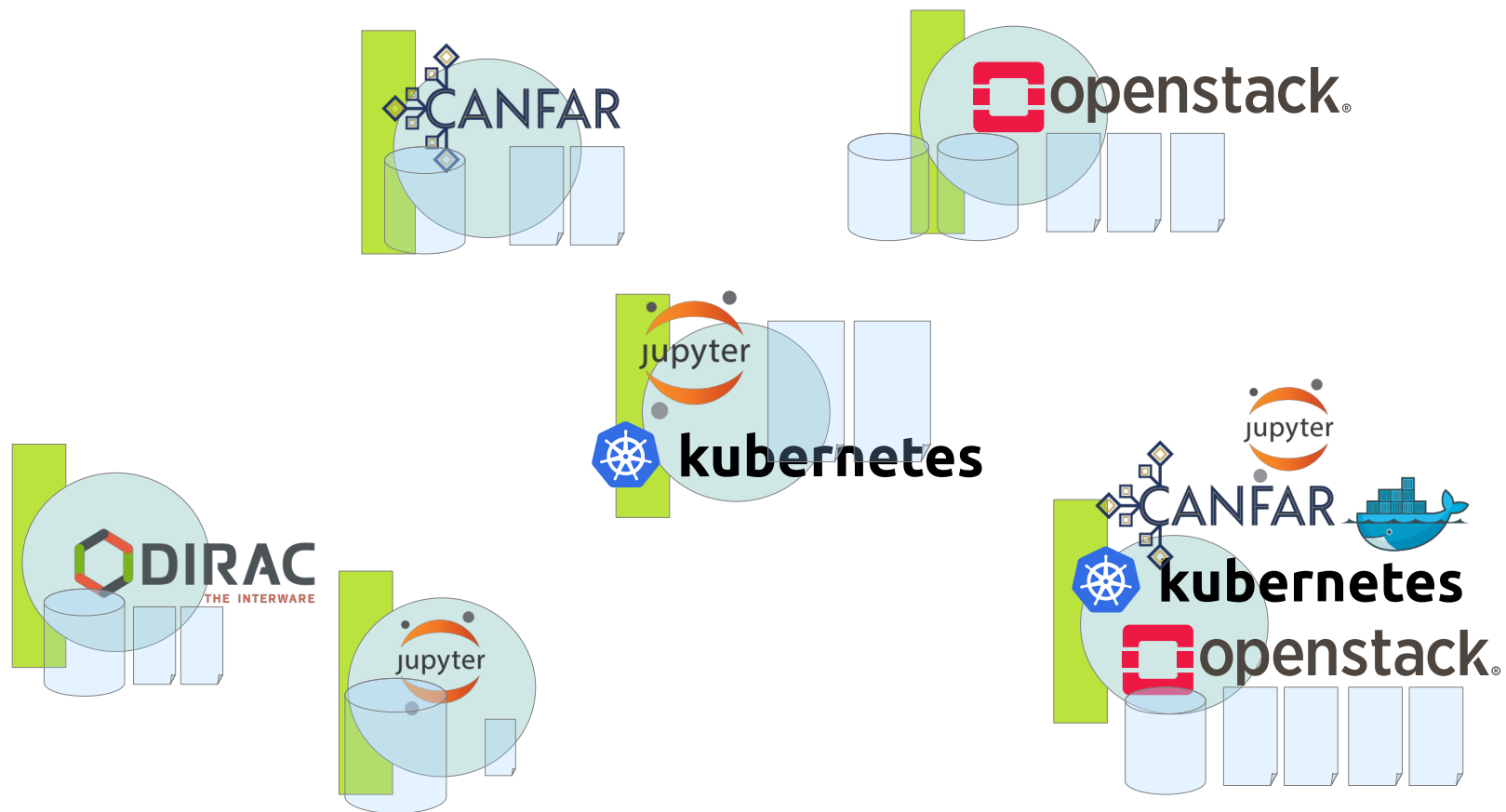
Lots of different execution platforms

Each with their own local capabilities and policies



Execution Broker – interface

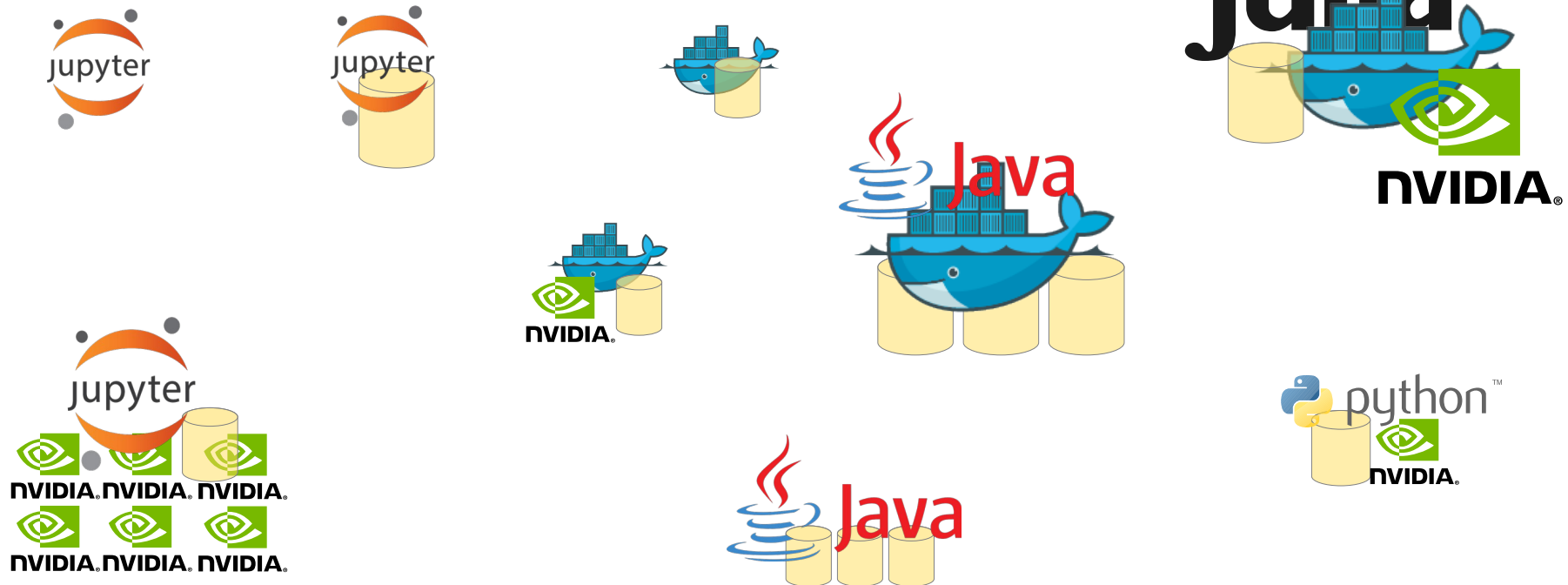
Deploys a common interface on each platform



The problem

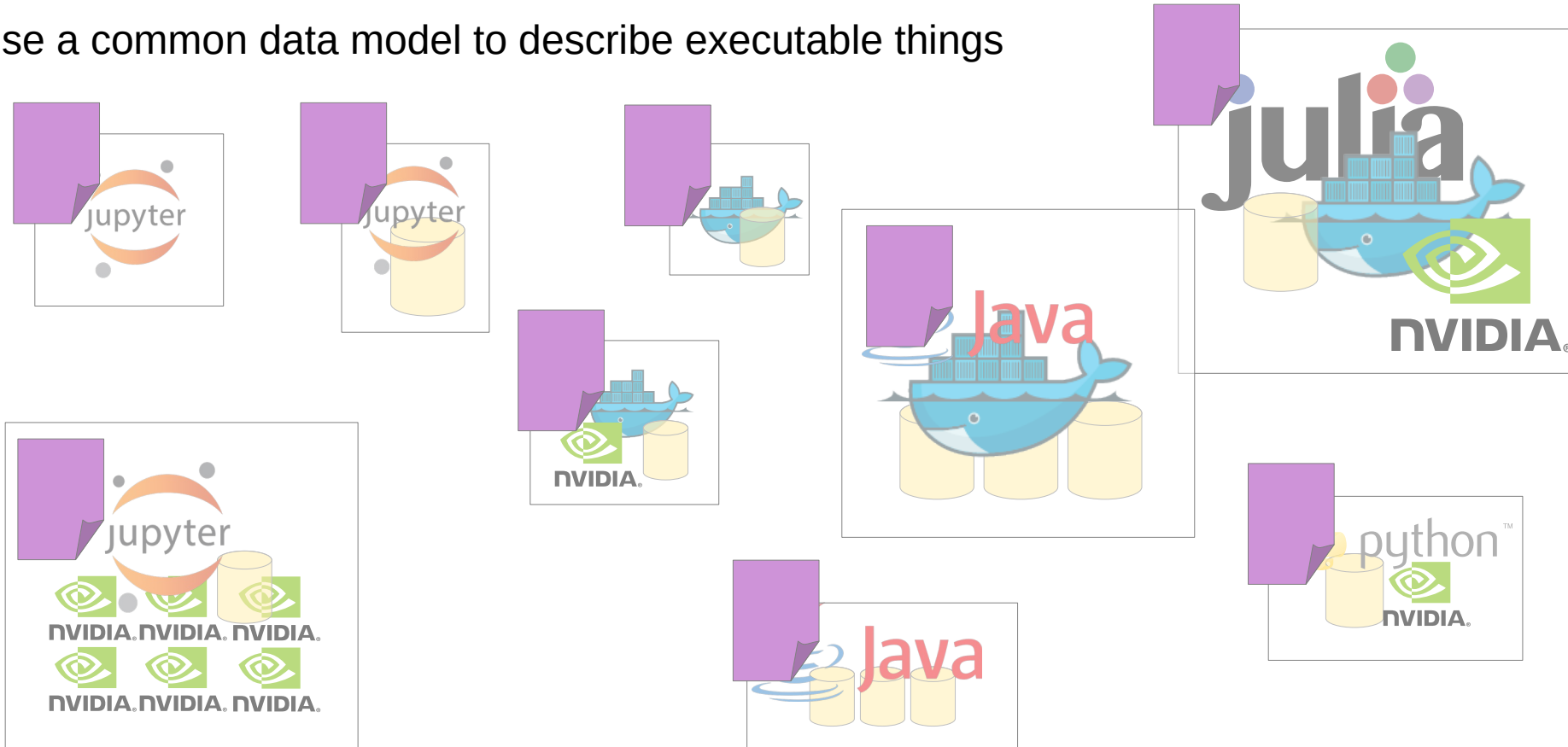
Lots of different types of software

Each with their own requirements and interfaces



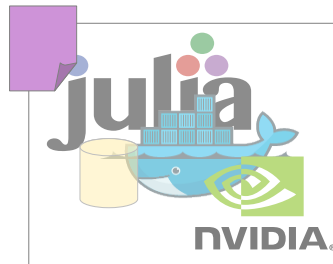
The Executable data model

Use a common data model to describe executable things

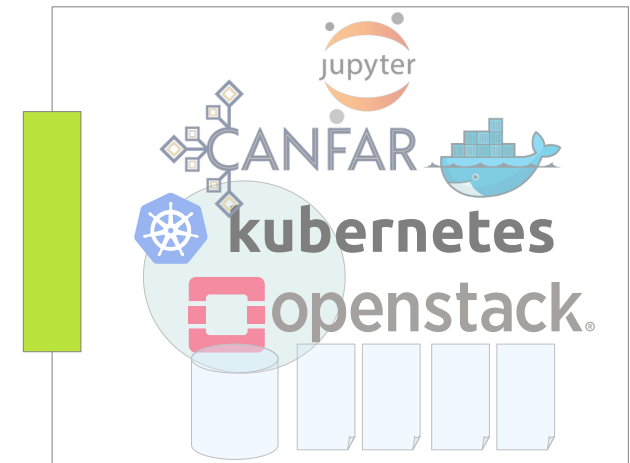
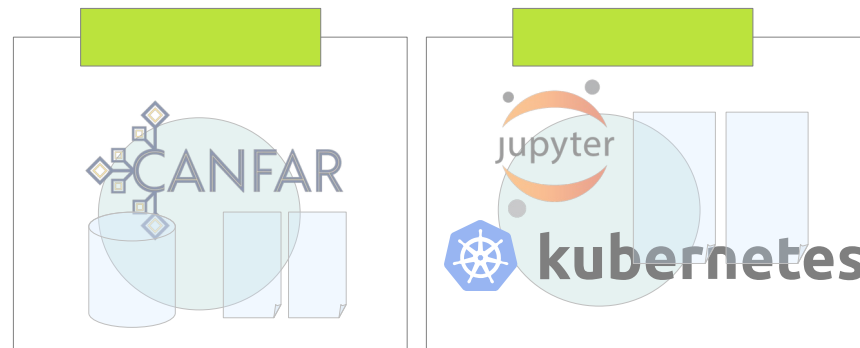
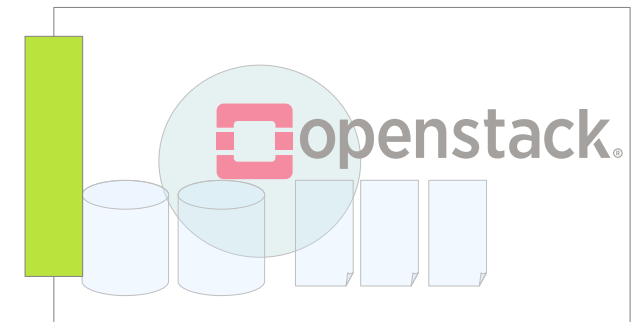


Execution Broker – the solution

Pass a common data-model description to a common interface

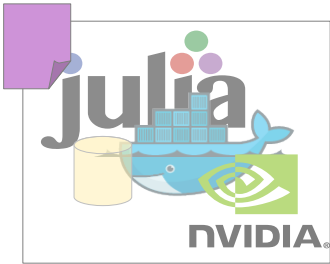


Can I run this ?

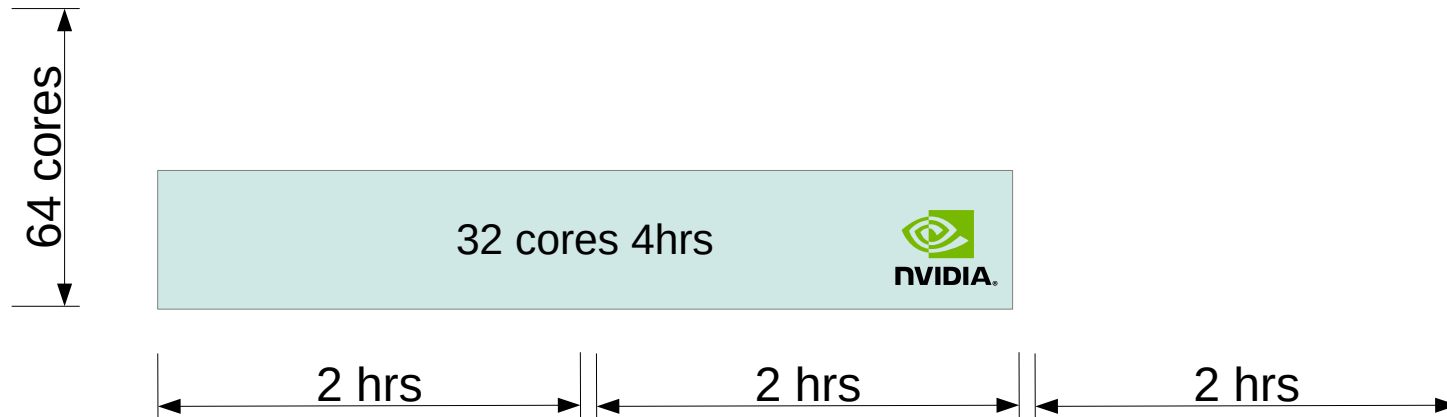


Resource scheduling

When can I run <this> ?

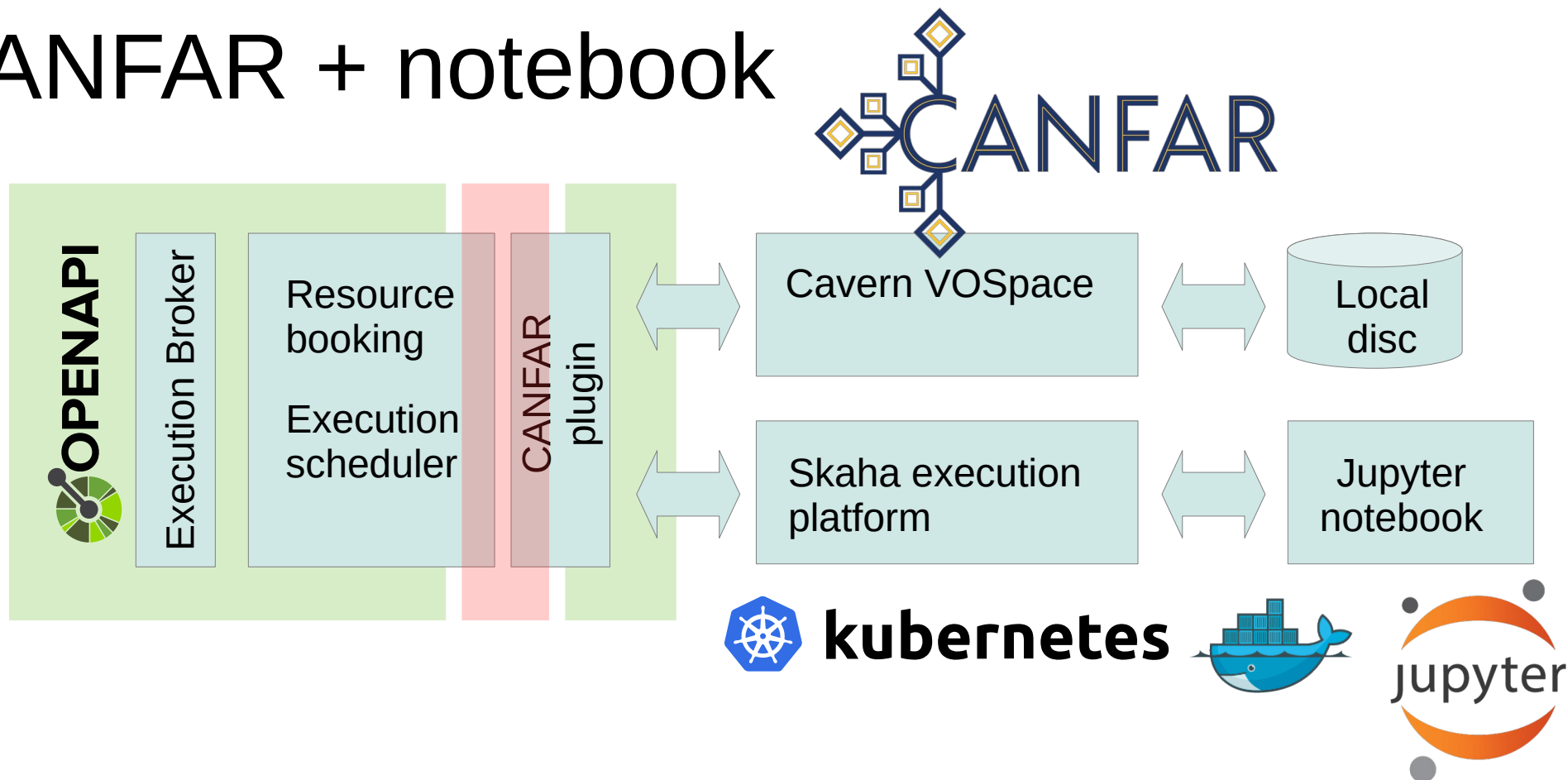


Request for 32 cores and a GPU for 4 hrs



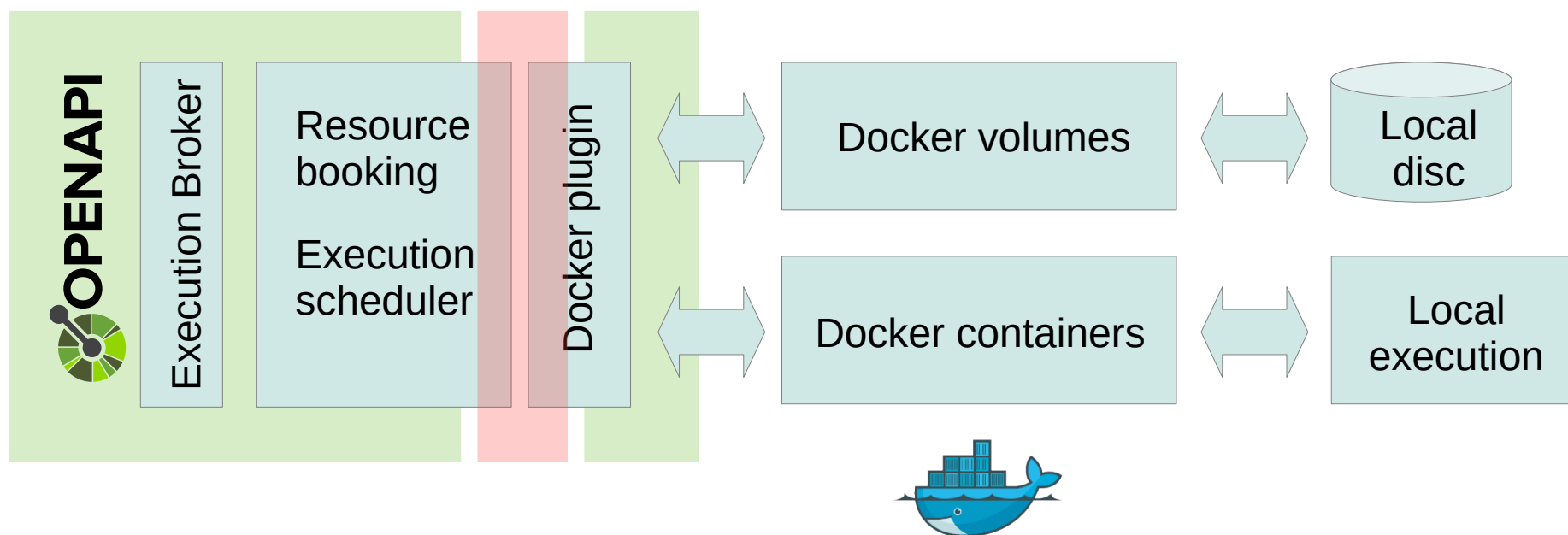
Prototype #1

CANFAR + notebook



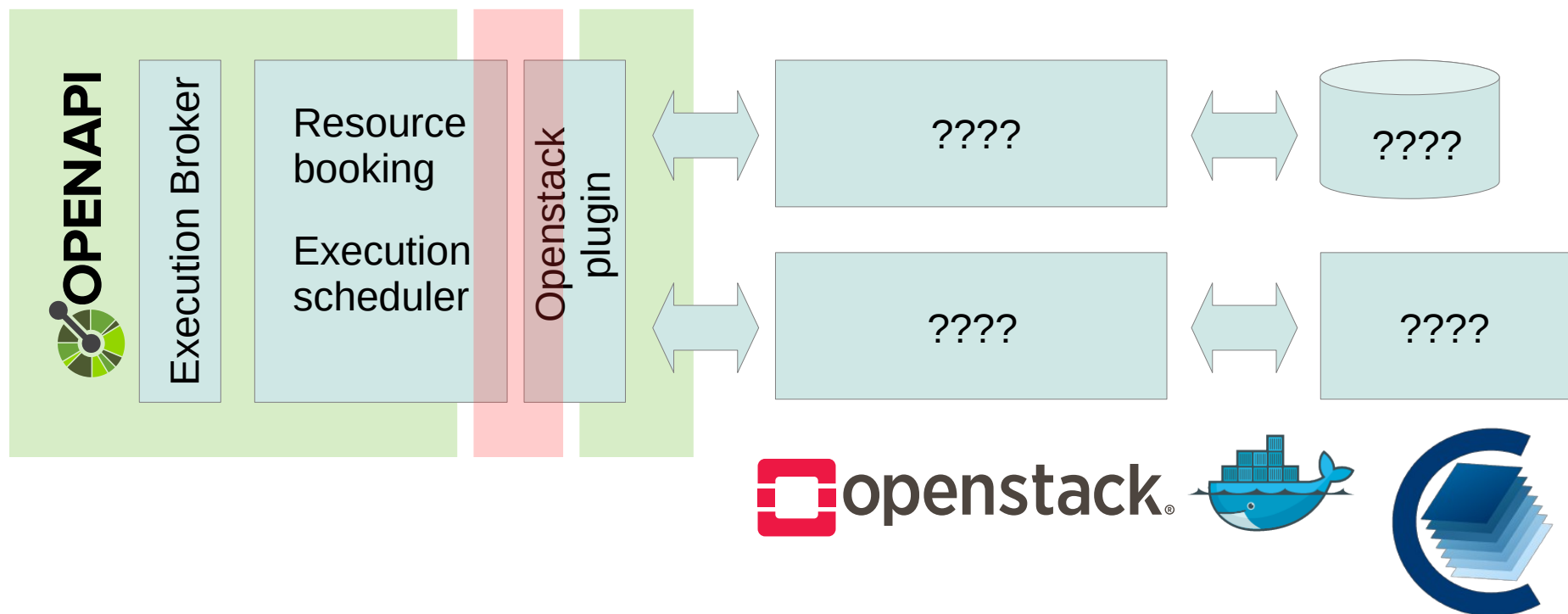
Prototype #2

Local Docker API



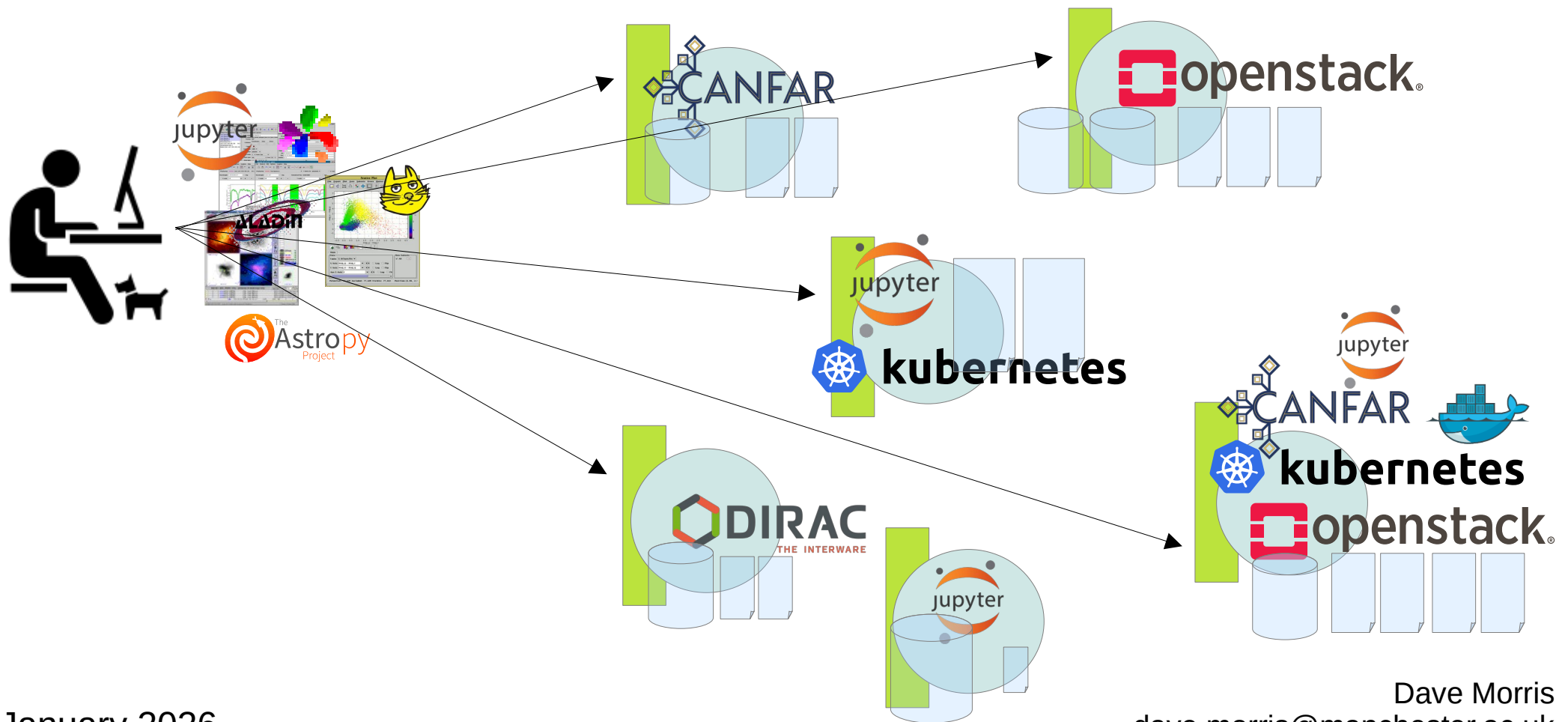
Prototype #3

???? + eMerlin data



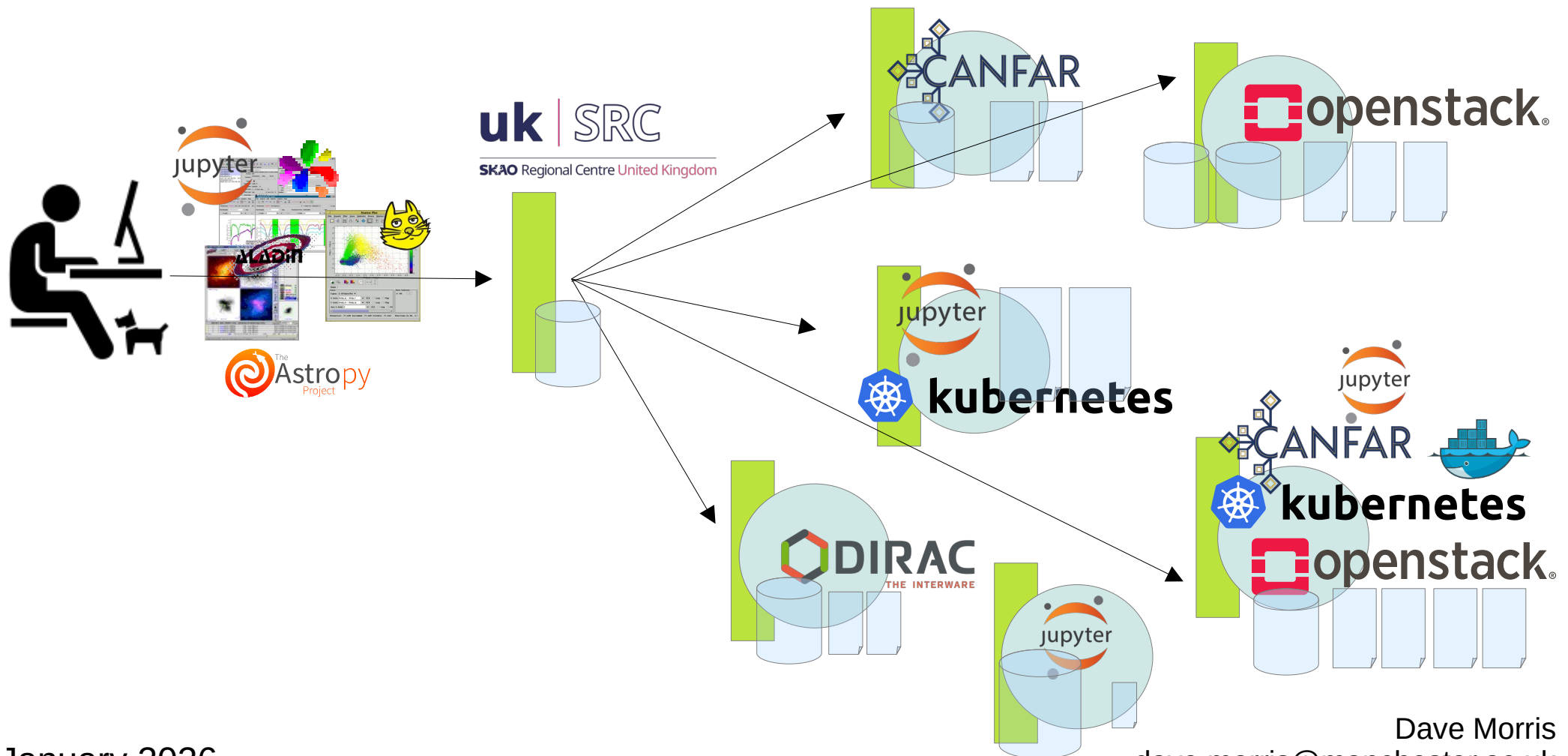
Execution Broker – user interface

How do we want to connect to the platforms ?



Execution Broker – aggregator service

How do we want to connect to the platforms ?



Prototype #n

Use cases

<your science use case here>

GPU **Rucio data** **Workflow**
FPGA **SrcNet data** **S3 data**
Slurm **Image processing pipeline**
Kubernetes **HTCondor**

Thank you

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<https://github.com/ivoa-std/ExecutionBroker>